Beyond SAS Programming:

SAS as a Coding Language AND Analytics Platform

SAS Innovate 2026 | Pre-Conference Educator Workshop

Lincoln H. Groves, PhD | Cristina Anton | SAS Academic Programs

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Why This Conversation Matters (Right Now)

Code is no longer the bottleneck

Anyone can generate code instantly
But employers don't just need syntax…

They need judgment

So how do we teach that?

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Why we care about this topic

Lincoln H. Groves, PhD

Work in SAS Academic Programs

Help universities modernize analytics education

From Does the code Run?

to Does it Live?

Cristina Anton

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Poll time: Who are you?

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Poll 1

What is your level of programming knowledge?

a) What code?

b) I've programmed before – but I don't remember much

c) I'm a straight up hacker

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Poll 2

What are your analytical tools of choice?

a) Excel

b) R

c) Python

d) SAS

e) Other

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Poll 3

Do you need to code to be a “real” data analyst?

a) Yes

b) No

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Let’s Pause…

What do your students believe makes a “good analyst”?

Let’s test those assumptions

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SAS as a Coding Language

[yes… we've been around for a long time]

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There are many paths in analytics.

But let's start with the code. SAS Code.

The **SAS Institute, Inc** is turning 50 this year.

And when some people think "SAS" they typically think "legacy code".

50 years will do that!

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More about SAS as a Coding Language

Why SAS Started With Code

- ✓SAS was one of the first higher-level languages
- ✓Built to make statistics accessible
- ✓Declarative, auditable, production-ready

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Introduce the OTJ Experience

Your Role Today: Analyst at iLink Mortgage

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Your Role Today: Analyst at iLink Mortgage

What You're About to Experience

- You're not starting from scratch…
  - You're inheriting someone else's work
- You then build upon the analysis by:
  - reviewing code
  - exploring data
  - comparing models
  - evaluating decisions

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SAS Code in Action

Demo | Part 1 | SAS Studio

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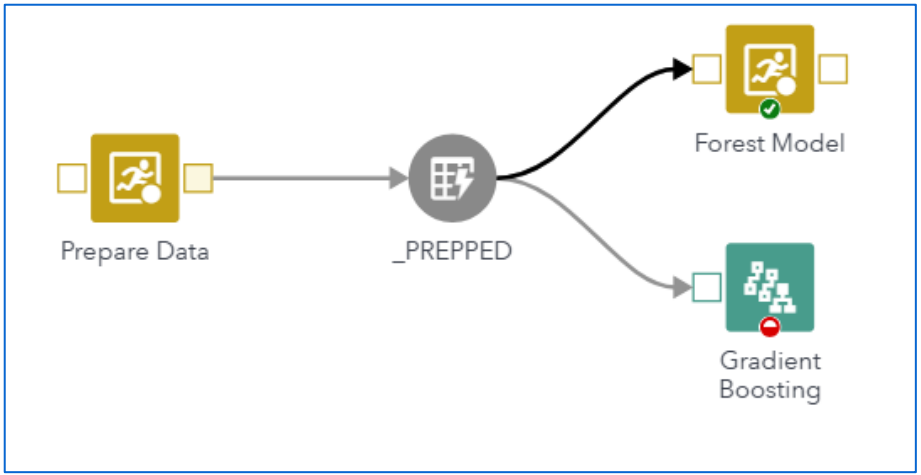
Demo 1 Prep | SAS Studio

What to pay attention to

Where do you have control?

What requires manual effort?

What breaks if you walk away?



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Demo 1 Debrief

SAS® Studio - Develop Code and Flows

New Options View Open Save All

SAS Studio compute context

Snippets

Filter

SAS Snippets My Snippets

Name

- Standard
- Viya Foundation
  - Cloud Analytic Services
  - Image Processing
  - Machine Learning
    - Compare Several ML Algorithms
    - Compare Two ML Algorithms
    - Generalized Linear Models
    - Load Data
    - Prepare and Explore Data
    - Supervised Learning
    - Unsupervised Learning

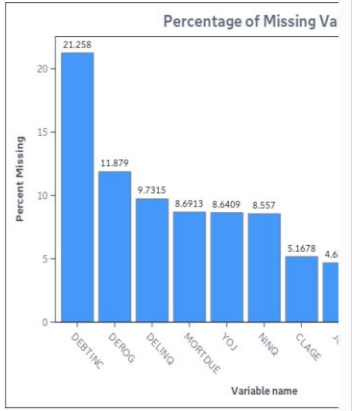
Start Page | Supervised Learning.snippet | * SAS Program.sas

Run Cancel Clear Copy to flow Snippet

Code

```
/* --Supervised Learning-- */

/* This snippet showcases a sample Machine Learning workflow for */
/* supervised learning using SAMPLEML.HMEQ data set. The steps include: */

/* (1) PREPARE AND EXPLORE */
/*     a) Load data set into CAS */
/*     b) Explore */
/*     c) Partition */
/*     d) Impute */
/*     e) Identify variables that explain variance */
/* (2) PERFORM SUPERVISED LEARNING */
/*     a) Fit model using random forest */
/* (3) EVALUATE AND IMPLEMENT */
/*     a) Score the data */
/*     b) Assess model performance */
/*     c) Generate ROC and Lift charts */

/* Define the macro variables for later use in the program */
/* Specify a folder path to write the temporary output files */
%let outdir = &_SASWORKINGDIR;

/* Create a CAS engine libref to save the output data sets */
%let caslibname = mycas;
libname &caslibname cas caslib=casuser;

/* Specify the data set names */
%let sasdata          = sampsio.hmeq;
%let casdata          = &caslibname..hmeq;
%let partitioned_data = &caslibname.._part;

/* Specify the data set inputs and target */
%let class_inputs     = reason job derog delinq ninq IM_clno IM_yoj;
%let interval_inputs = loan value IM_clage IM_debtinc IM_mortdue;
%let target           = bad;
```

Log | Results | Output Data (13)



What does code give you?

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Does it Work? vs Can it Live?

[deeper dive]

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The Enterprise Questions(s)

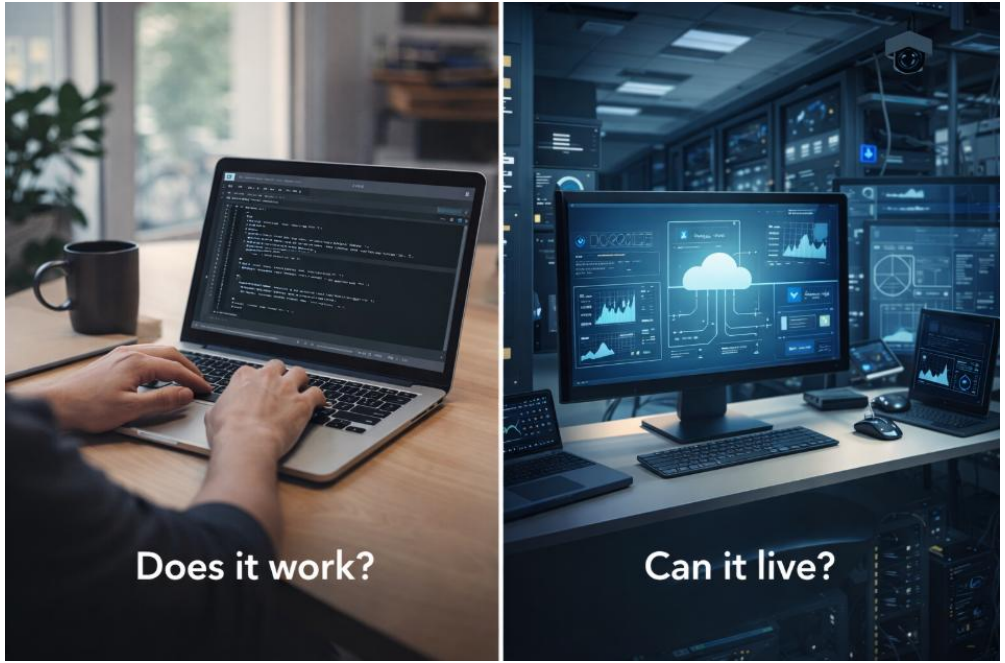
What happens when we try to put this model into production?

What if this model needs to be:

- ✓Reused
- ✓Governed
- ✓Deployed
- ✓Monitored?

Code answers: “Does it work?”

Enterprise asks: “Can it live?”



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SAS as an Analytics Platform

[*Does it Work?* v. *Can it Live?*, in action]

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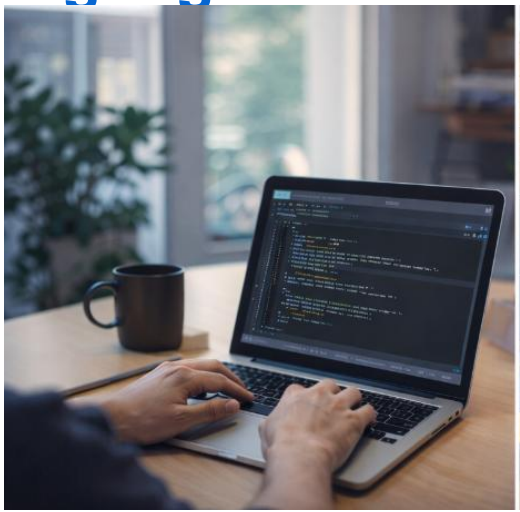
SAS as Two Things

SAS Programming Language

SAS as an Analytics Platform



Syntax
Control
Explicit logic
Reproducibility

Orchestration
Collaboration
Governance
Deployment

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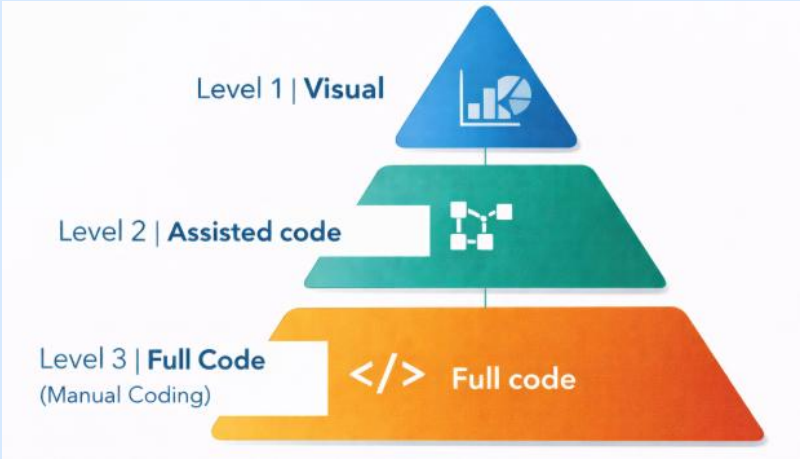
Levels of Abstraction from the Code

No longer a “Code vs No-Code” debate

Analytics in levels



Moving up increases orchestration.

These are not competitors.

They are layers of thinking.

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Level 1: The No-Code Path

Demo | Part 2 | SAS Visual Analytics

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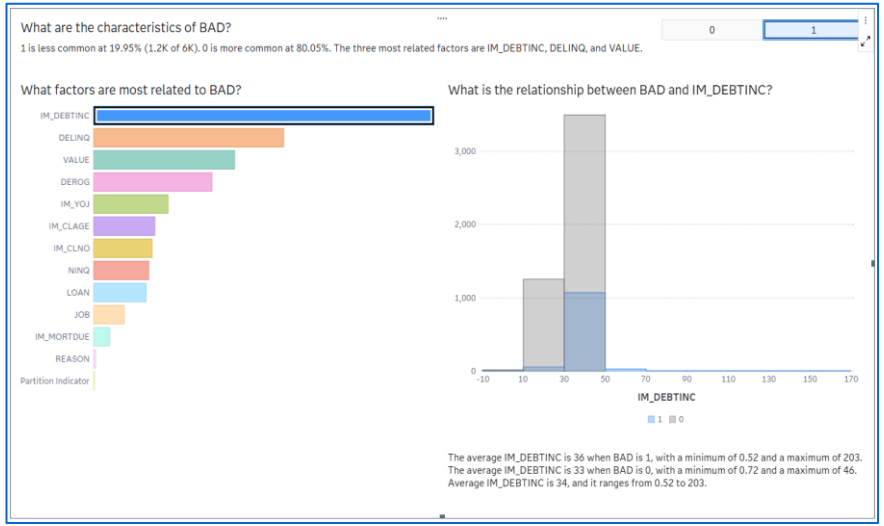
Demo 2 Prep | SAS Visual Analytics

Reminder of what to pay attention to



Where do you have control?

What requires manual effort?

What breaks if you walk away?

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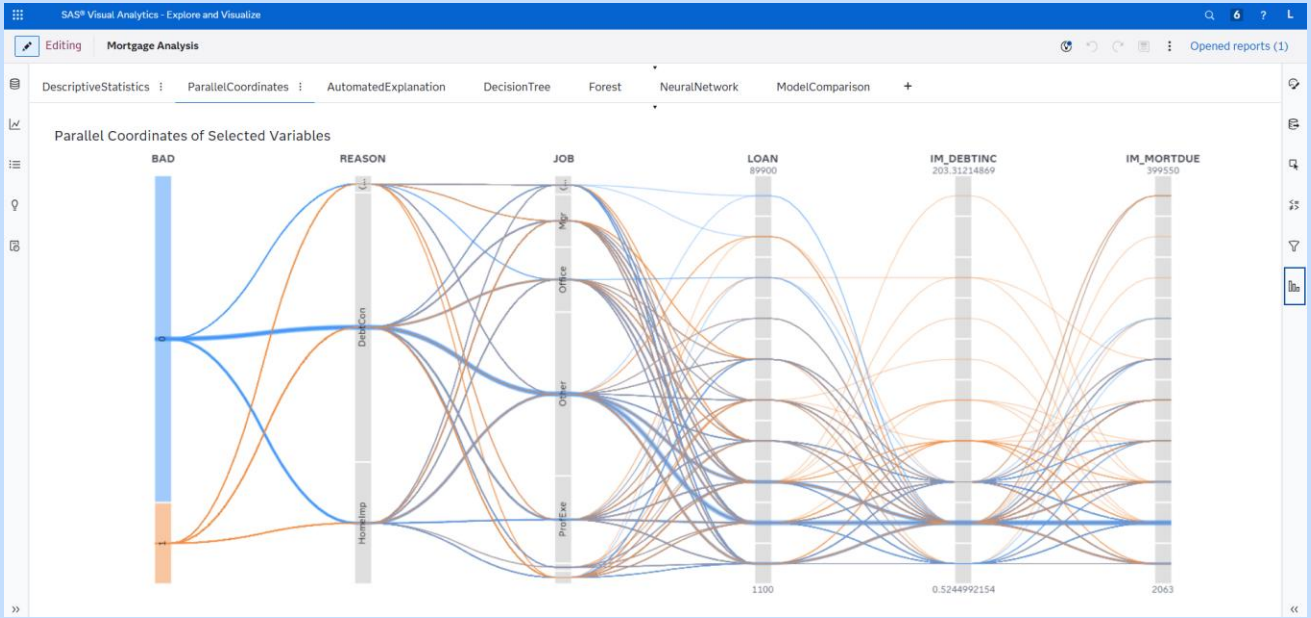
Demo 2 Debrief



What does SAS Visual Analytics give you?

What just got easier?

What just got abstracted?

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Level 2: The No/Low-Code Path

Demo| Part 3 | SAS Model Studio

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Demo 3 Prep | SAS Model Studio

Reminder of what to pay attention to

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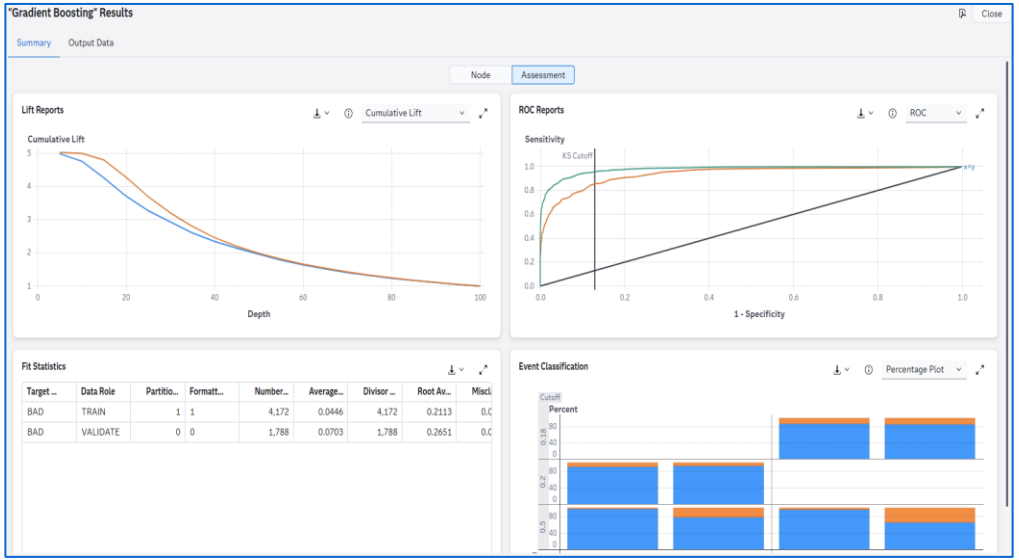

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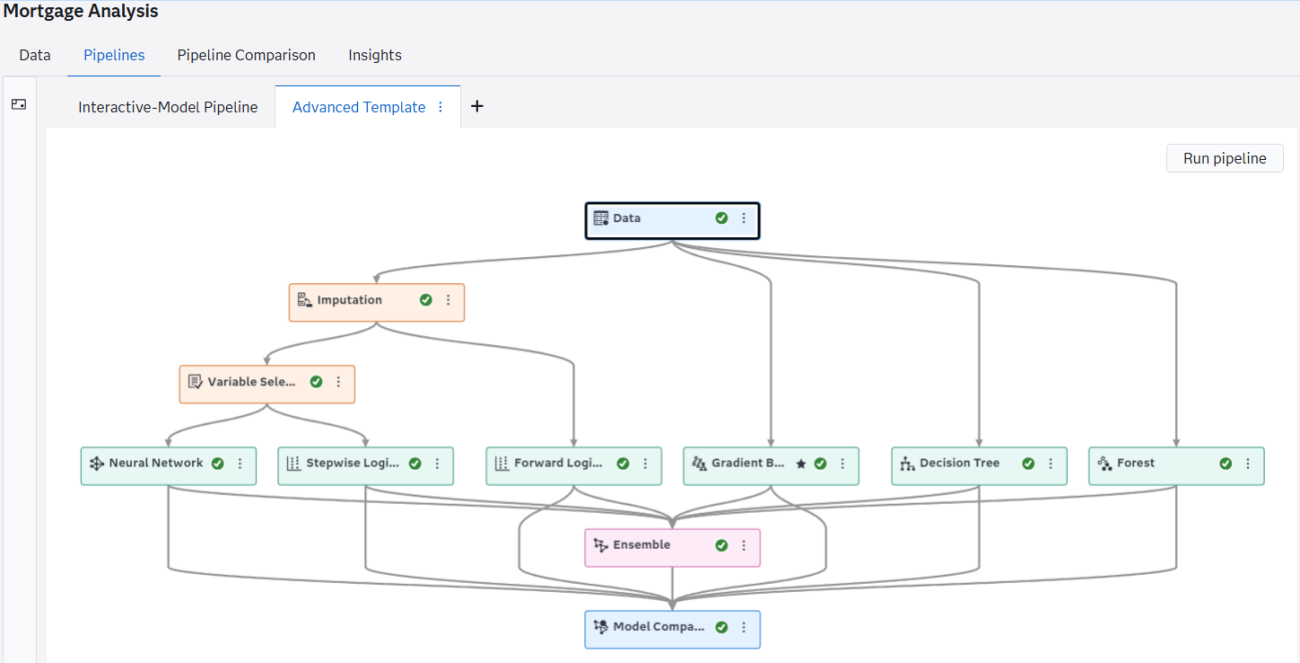
Demo 3 Debrief



What does SAS Model Studio give us?

What just got easier?

What just got abstracted?

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Some (Project) Takeaways

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Enterprise Analytics Requires Multiple Modalities

Need	Code	Platform
Speed	It depends on complexity	Strong
Transparency	Strong	It depends on configuration
Governance	Strong	Strong
Collaboration	It depends on workflow	Strong
Deployment	It depends on tooling	Strong

Enterprise systems are not built at one level of abstraction.

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How People Actually Learn Analytics

Learning rarely happens in a straight line

From beginner… to pro… the path:

- ✓Start visual → curiosity grows
- ✓Peek under the hood → code appears
- ✓Go deep in code → return to visuals for communication

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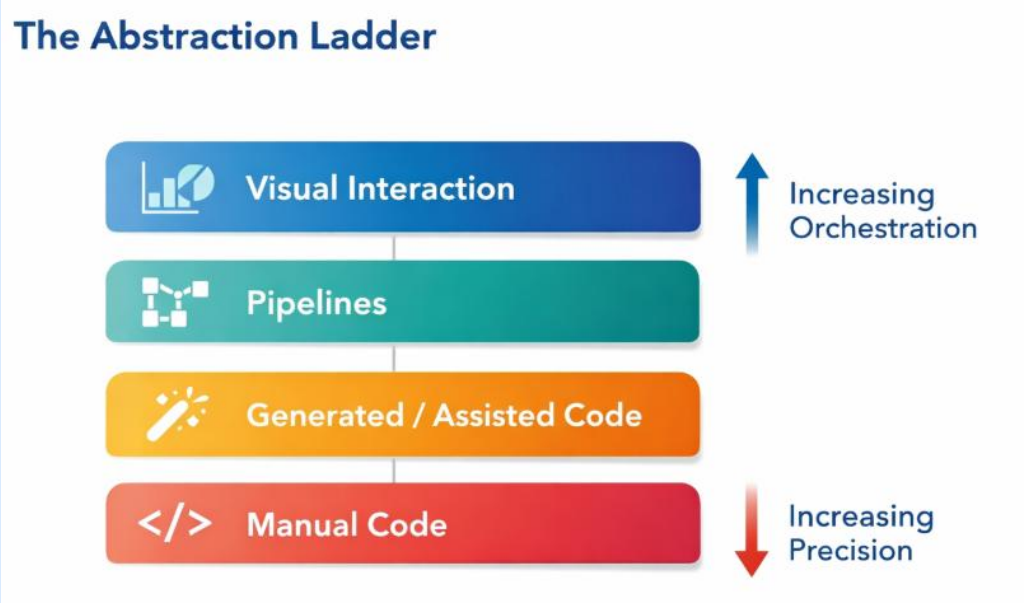
The Big Idea

Why This Matters for You, Part 1

Analytics maturity isn't just about the tool.

It's about knowing when to raise or lower abstraction from the code.



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The Big Idea

Why This Matters for You, Part 2

- ✓ Analytics careers will not live in one interface.
- ✓ Enterprises expect governance literacy.
- ✓ Abstraction fluency is a differentiator.
- ✓ AI will accelerate this shift.

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But… one parting thought

AI is collapsing the need to *stay* at one level of abstraction

How?

- ✓ Code generation
- ✓ Pipeline explanation
- ✓ Model interpretation

Implications

The **Code versus No code** debate was already fading.

AI will accelerate it.

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What This Means for Teaching

The BIG recap

What Students Learn Today	What They Actually Need
Build a model	Compare + defend models
Write code	Navigate tools + workflows
Optimize accuracy	Balance performance + responsibility

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Helpful Resources

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My Team: Global Academic Programs

Our Mission: Empowering Analytics for Academics

We support the next generation by providing free software and resources for academia.

Our “customers” are students, educators, independent learners, and academic researchers.

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Resources for Teaching and Learning

SOFTWARE

- SAS® Viya® for Learners
- SAS® Viya® Workbench for Learners

TRAINING

- e-Learning
- How-to videos
- Onsite & virtual training

EDUCATIONAL RESOURCES

- Teaching materials
- Academic specializations
- SAS Educator Portal
- SAS Skill Builder for Students

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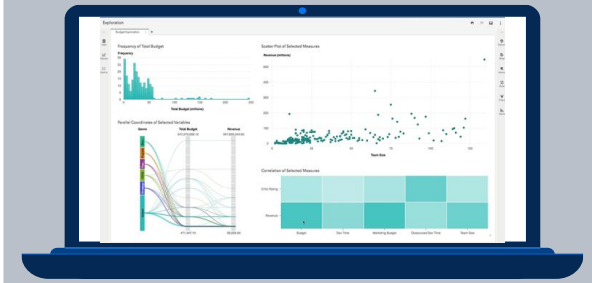
Software for Teaching and Learning

Free Academic Software in the Cloud

VFL

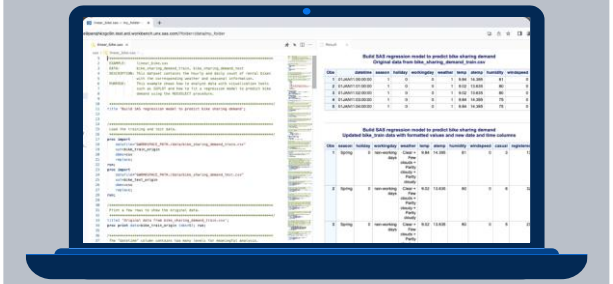
SAS® Viya® for Learners



www.sas.com/viyaforlearners

WFL

SAS® Viya® Workbench for Learners



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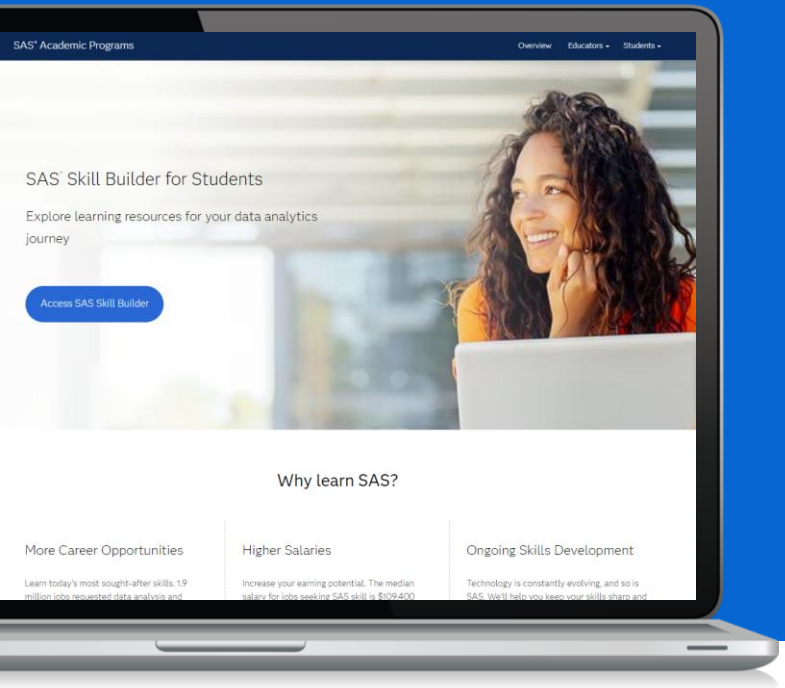
SAS Skill Builder for Students

Your (analytical) future continues here



- Learn on your time
  - Free e-learning courses
  - Certification prep materials
- Free software
- Career resources
  - Discover why data analytics is a rewarding career
  - Learn about what it means to be a data analytics professional
  - Connect with employers
- Cross-industry access
  - Analytical skills for any discipline

www.sas.com/skillbuilder

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SAS Innovate 2026

More opportunities to attend events featuring SAS Academic Programs!

Hackathon

- SAS Hackathon Boot Camp for Academics | Tues | 1:30

Networking

- Faculty Connect | Today | Welcome Reception @ Education Booth
- Student Braindates | Tue | 10am
- Networking Lunch | Tue | 11:15am
- Student Showcase | Tue | 6pm

Workshops

- You + Me + ChatGPT Make Three: Leveraging ChatGPT to get started with SAS Coding | Wed | 8:30am
- Dashboarding: Where Data becomes Insights | Wed | 9:30a
- Ethical Data Analysis using SAS Viya | Wed | 2:45pm
- The Polyglot Data Scientist: SAS® , Python, & R in SAS® Viya® Workbench | Wed | 3:45pm
- Predict the Next "Movie Classic" Blockbuster: Text Analytics in SAS® Model Studio | Thur | 9:00am

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Thank you!

Connect with us on LinkedIn:

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